

**THE WORD GARDEN:
EVALUATING MULTIMODAL MULTIPLAYER GAMES FOR
COGNITIVE AND SOCIAL SKILLS DEVELOPMENT
IN DYSLEXIC CHILDREN**

25-26J-421

Final Project Report

BSc (Hons) Degree in Information Technology Specialized in Interactive Media

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

DECLARATION

We declare that this is our own work, and this report does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning. To the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Children with dyslexia face not only persistent literacy challenges but also heightened anxiety, diminished motivation, and social withdrawal arising from repeated failure in educational environments built around individual text-based interaction. Existing digital interventions address these children in isolation, through single-player, single-device tools that prioritise phonological accuracy while ignoring the social, emotional, and motivational dimensions equally critical to sustained engagement and holistic development. This report presents The Word Garden, a cooperative multiplayer educational game for dyslexic children aged 8 to 12, developed and evaluated as an integrated dual-device system combining a 3D PC game environment with a paired Android companion application.

The Word Garden is the first platform to combine synchronous cooperative multiplayer gameplay, cross-device multimodal literacy tasks, voice-enabled peer communication, anticipatory feedback, and a non-competitive individual reward system within a single dyslexia-focused educational game. Children explore a shared 3D world on PC, discover letter chests cooperatively, and complete private letter tracing tasks on their paired Android device, keeping individual practice hidden from peers to eliminate performance anxiety while maintaining social connection through shared narrative progression and real-time voice coordination via Steam overlay. The system is designed to address dyslexia's cognitive, emotional, and social barriers simultaneously rather than remediating isolated skills in isolation.

The platform is investigated across four integrated research dimensions. The Multimodal Gameplay Module evaluates whether the cooperative multiplayer environment produces measurable improvements in attention, task performance, and retention compared to a matched single-player condition. The Social Engagement System investigates the impact of voice-enabled peer communication and a non-competitive individual reward framework on motivation, persistence, and social confidence. The Predictive Feedback Framework examines whether anticipatory multimodal feedback delivered before likely errors improves accuracy, retention, and motivation compared to conventional post-error correction. The Cognitive

Interaction Engine investigates cognitive load effects of the integrated motor-cognitive gameplay mechanics within the multiplayer context.

Technical evaluation over three weeks with 2 to 5 concurrent players demonstrated system-wide stability: frame rates of 70 to 80 FPS on discrete GPU hardware, network latency scaling from 60ms at two players to 150ms at five players, PC-to-smartphone event delivery accuracy exceeding 98.3% via Firebase Realtime Database middleware, and zero crashes across 48 hours of continuous testing. Expert validation by two consultant child and adolescent psychiatrists from Lady Ridgeway Hospital confirmed alignment between the integrated platform design and therapeutic principles for dyslexia intervention, with Cooperative Mechanics and Multisensory Integration each receiving perfect ratings of 5.0 out of 5.0. Pilot evaluation was conducted at a private English language class with Grade 1 and Grade 2 students following dyslexia screening using a clinically validated instrument. Preliminary findings confirmed positive engagement, reduced performance anxiety, and meaningful peer interaction across all observed sessions.

Keywords:

Dyslexia, Cooperative Multiplayer, Educational Game, Dual-Device Learning, 3D Letter Interaction, Voice Communication, Reward System, Anticipatory Feedback, Cognitive Load, Serious Games, Interactive Media, Unreal Engine, Word Garden, Inclusive Design

ACKNOWLEDGEMENT

The successful completion of The Word Garden as an integrated research system reflects the collective effort, guidance, and generosity of many individuals and institutions. We are sincerely grateful to each of them.

We express our deepest appreciation to our supervisor, Mr. Didula Chamara Thanaweera Arachchi, and co-supervisor, Mr. Nushkan Nizme, of the Sri Lanka Institute of Information Technology, for their consistent mentorship, critical engagement, and encouragement across every phase of this research. We are equally grateful to Dr. Kasun Karunanayaka of the University of Colombo School of Computing for his facilitation and broader academic guidance.

We extend sincere thanks to Dr. Swarna Wijetunga, Consultant Child and Adolescent Psychiatrist and Founder of the Sri Lanka College of Child and Adolescent Psychiatrists, and Dr. Darshani Buddhika Hettiarachchi, Consultant Child and Adolescent Psychiatrist at Lady Ridgeway Hospital, for their invaluable clinical insights during expert evaluation and for the dyslexia screening instrument used in our pilot evaluation. Their professional oversight was indispensable in ensuring the platform remained clinically grounded, therapeutically appropriate, and ethically sound throughout.

We are deeply grateful to the management, educators, and staff of the private English language class that facilitated our pilot testing sessions, and to all participating children and their parents, whose willingness and trust made meaningful evaluation possible.

Finally, we thank the Faculty of Computing at SLIIT for the institutional and ethical framework within which this research was conducted, and our families and friends for their patience and unwavering support throughout this journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADHD	Attention Deficit Hyperactivity Disorder
API	Application Programming Interface
DB	Database
FPS	Frames Per Second
HUD	Heads-Up Display
JSON	JavaScript Object Notation
LRH	Lady Ridgeway Hospital
NAT	Network Address Translation
P2P	Peer-to-Peer
PC	Personal Computer
REST	Representational State Transfer
RPC	Remote Procedure Call
SBSCH	Sirimavo Bandaranaike Specialized Children's Hospital
SDLC	Software Development Life Cycle
SDK	Software Development Kit
SLIIT	Sri Lanka Institute of Information Technology
UCSC	University of Colombo School of Computing
UI	User Interface
UMG	Unreal Motion Graphics
VOIP	Voice Over Internet Protocol

1. INTRODUCTION

1.1 Background

Dyslexia, a neurobiological learning disorder affecting approximately 5 to 10 percent of children globally, is characterised by persistent difficulties in word recognition, spelling, and phonological decoding despite average or above-average intelligence [1], [2]. Beyond these cognitive challenges, dyslexic learners consistently experience reduced academic self-concept, heightened anxiety around literacy tasks, and social withdrawal from group learning activities [7], [19]. In many contexts including Sri Lanka, access to specialised dyslexia support remains limited, leaving a significant proportion of children without effective intervention [1].

The landscape of existing digital dyslexia interventions, while growing, is defined by a common set of limitations: single-player formats that isolate learners from peer support, flat 2D text presentations that amplify visual processing difficulties, accuracy-based metrics that reinforce fear of failure, and reactive post-error feedback that corrects after mistakes rather than guiding before them [4], [19]. No existing digital platform had combined cooperative multiplayer gameplay, cross-device multimodal interaction, voice-enabled peer communication, anticipatory feedback, and a non-competitive individual reward system within a single dyslexia-focused educational experience prior to this research.

The Word Garden was created in direct response to these limitations. It is a cooperative multiplayer educational game in which children with dyslexia learn together rather than alone, addressing the cognitive, emotional, and social barriers of dyslexia simultaneously within a single integrated system. Children explore a shared 3D world on PC, discover letter-containing chests cooperatively, and complete private letter tracing tasks on a paired Android device — keeping individual practice hidden from peers to eliminate performance anxiety while maintaining full social connection through shared gameplay and real-time voice communication via Steam overlay.

1.2 The Word Garden System

The Word Garden is built on Unreal Engine 5.4 for PC-based gameplay and integrated with a Flutter-based Android companion application. Firebase Realtime Database serves as cloud-based middleware synchronising events between devices in real time. Steam Online Subsystem manages multiplayer session discovery, world state replication, and voice communication through a listen-server topology supporting 2 to 5 concurrent players connected via four-digit session codes. The system employs a dual-device architecture in which the PC game manages cooperative world exploration and social interaction while the Android application delivers private multisensory literacy tasks.

When players discover letter chests in the shared 3D environment, a Firebase event triggers a private letter tracing task on the paired mobile device. The child completes the task at their chosen difficulty level — guided tracing with full path support, reduced-guidance tracing, or recognition-only selection — and only the completion status is returned to the PC, ensuring individual literacy performance is never visible to peers. Successfully completed tasks unlock the chest and add the letter to the shared inventory. Players use Steam voice to coordinate letter arrangement at the central island, collaboratively form target words, and receive individual motivational rewards through a non-competitive system of badges, stars, and spoken praise that acknowledges effort and participation rather than accuracy or speed.

Throughout gameplay, the system delivers anticipatory multimodal feedback — visual glowing cues, subtle audio tone shifts, and haptic pulses — when trajectory patterns suggest a likely error is approaching, guiding learners proactively before mistakes occur rather than correcting after failure. This anticipatory mechanism represents a fundamental departure from the reactive feedback paradigm dominant in existing educational tools.

System Feature	Description
Cooperative Multiplayer	2 to 5 children explore a shared 3D world, discover letter chests, and form words together
3D Letter Interaction	Letters presented as interactive 3D objects replacing flat 2D text throughout the game world
Dual-Device Architecture	PC handles social exploration; Android app delivers private literacy tasks simultaneously
Real-Time Synchronisation	Firebase middleware syncs chest discovery events and task completion between devices (<200ms)

System Feature	Description
Voice Communication	Steam overlay P2P voice, opt-in, non-proximity-based, enabling free peer coordination
Anticipatory Feedback	Visual, audio, and haptic cues before likely errors occur on mobile tracing tasks
Non-Competitive Rewards	Individual badges, stars, and spoken praise rewarding effort and participation, no leaderboards
Adaptive Difficulty	Three tracing levels: guided, reduced-guidance, and recognition-only with visual theme options
Timestamped Event Logging	All gameplay actions logged with timestamps for rigorous research evaluation

Table 1: Core Features of The Word Garden Integrated System

1.3 Research Dimensions

The Word Garden is investigated across four integrated research dimensions, each contributing empirical evidence on a distinct aspect of the unified system's effectiveness. These dimensions are not separate systems but interlocking layers of the same game evaluated from different analytical perspectives.

The Multimodal Gameplay Module investigates whether the cooperative multiplayer environment produces measurable improvements in attention, task performance, and retention in dyslexic children compared to a matched single-player condition using the same game world and task content. The Social Engagement System investigates how voice-enabled peer communication and the non-competitive individual reward framework affect motivation, persistence, and social confidence within the multiplayer context. The Predictive Feedback Framework investigates whether the anticipatory multimodal feedback mechanism reduces frustration and improves accuracy and motivation compared to conventional post-error feedback during mobile tracing tasks. The Cognitive Interaction Engine investigates the cognitive load experienced by players during the integrated multiplayer gameplay, comparing subjective difficulty ratings across sessions using a child-appropriate Faces Scale instrument.

Together these four dimensions generate the first controlled empirical evidence base for a cooperative, voice-enabled, cross-device multiplayer educational game designed specifically for dyslexic children — filling a gap confirmed across multiple systematic reviews and

generating design knowledge applicable to the next generation of inclusive educational technology.

2. BACKGROUND AND LITERATURE SURVEY

2.1 Dyslexia: Cognitive and Socio-Emotional Dimensions

Dyslexia is a neurobiological learning disorder characterised by persistent difficulties in phonological processing, word recognition, and spelling despite adequate intelligence and educational opportunity, affecting 5 to 10 percent of children worldwide [1], [2]. The double-deficit hypothesis identifies deficits in both phonological processing and rapid automatised naming as the primary impairments [3], while neuroimaging research has demonstrated atypical left-hemisphere reading network activation that normalises following targeted intervention [5]. The cerebellar deficit hypothesis further proposes that deficits in procedural learning and motor automaticity contribute to the literacy and spelling difficulties observed in dyslexic learners [6].

Beyond cognitive manifestations, dyslexia consistently correlates with reduced academic self-concept, heightened anxiety around literacy tasks, fear of failure, and social withdrawal from group learning contexts [7], [19]. These socio-emotional consequences are frequently as damaging to long-term outcomes as the cognitive deficits themselves. Existing digital literacy tools for dyslexic children, despite providing some scalability advantages over specialist-led instruction, systematically fail to address these emotional and social dimensions. They isolate learners in single-player formats, present content in flat 2D text that amplifies visual confusion, and apply competitive accuracy-based metrics that reinforce rather than alleviate fear of failure [4], [19].

2.2 Game-Based and Multiplayer Learning

Game-based learning environments offer structural engagement mechanisms unavailable in conventional instruction: clear goal structures, immediate feedback, graduated challenge, narrative context, and social dynamics that together support sustained attention and intrinsic motivation [14], [19]. Gamified literacy interventions have demonstrated improvements in motivation, engagement, and reading outcomes for learners with dyslexia, confirming that game-based contexts can produce transferable cognitive benefits [9]. Meta-analytic evidence

confirms that digital game-based training produces measurable improvements in cognitive and literacy outcomes across children with neurodevelopmental disorders including dyslexia [10].

The social dimension of multiplayer contexts provides mechanisms that single-player equivalents cannot replicate: peer accountability, shared goal structures, social normalisation of difficulty, and collaborative problem-solving that keeps individual learners engaged through collective experience [13], [14]. Research on group gameplay for dyslexic children found that multiplayer contexts produced spontaneous peer scaffolding and increased social engagement, with the cooperative dynamic sustaining participation in ways that isolated single-player play did not [13]. For dyslexic learners who commonly experience social withdrawal and reduced confidence, structured multiplayer settings provide meaningful opportunities for peer interaction and social growth within a low-pressure virtual environment [7], [19].

2.3 Multisensory Learning and 3D Interaction

Multisensory learning approaches integrating visual, auditory, and kinesthetic modalities simultaneously strengthen phonological neural connections and improve decoding accuracy, with evidence that engaging multiple sensory channels reduces the extraneous cognitive load imposed by any single modality [11]. Presenting letters as interactive 3D objects rather than flat 2D text provides additional spatial cues that support letter discrimination and reduce the visual confusion that standard text layouts produce for dyslexic readers [11]. Kinesthetic reinforcement through letter tracing on mobile devices creates motor memory traces that complement visual letter recognition in the PC environment, establishing redundant encoding pathways that support retention in learners with working memory difficulties [6], [11].

The dual-device architecture of The Word Garden distributes multisensory engagement across devices without creating interface complexity on any single screen. The PC provides the immersive social and visual context while the Android device delivers the tactile-kinesthetic literacy task privately, enabling simultaneous multisensory engagement without cognitive interference between the modalities [15]. Adaptive difficulty levels and visual theme personalisation on the mobile application further reduce cognitive barriers by accommodating individual learner profiles [12].

2.4 Feedback Design for Dyslexic Learners

The dominant design pattern in educational games is reactive post-error feedback, which informs learners only after mistakes have been made. For dyslexic learners already prone to fear of failure and learned helplessness, repeated exposure to post-error correction can reinforce anxiety and progressive disengagement rather than supporting resilience [19]. Anticipatory feedback mechanisms that provide gentle guidance before likely errors occur represent a principled departure from this paradigm, with potential to reduce frustration and build proactive confidence during learning activities [20]. Research on intelligent tutoring systems confirms that feedback timing significantly affects learning outcomes and learner affect, with earlier guidance generally producing better error prevention and reduced frustration [22].

Multimodal anticipatory feedback combining visual, audio, and haptic channels provides redundant guidance signals that can reach learners through whichever sensory channel is most accessible in a given moment, reducing the processing demands of any single feedback modality and accommodating the diverse sensory processing profiles common across the dyslexia population [11].

2.5 Cognitive Load in Educational Game Contexts

Cognitive load theory distinguishes between intrinsic load from task complexity, extraneous load from poor interface design, and germane load associated with schema formation [27]. For dyslexic learners, conventional educational game interfaces impose high extraneous load through text-heavy instructions and accuracy-focused interfaces, consuming resources that would otherwise support learning [23]. The dual-device architecture of The Word Garden addresses extraneous load by separating social exploration from literacy task execution, reducing interface complexity on each screen while maintaining rich multimodal engagement across the system as a whole [15]. Measuring cognitive load through child-appropriate instruments, specifically a Faces Scale administered after each gameplay round rather than adult verbal report scales, enables accurate subjective workload assessment without introducing additional reading demands for dyslexic participants [26].

2.6 Voice Communication in Educational Contexts

Voice-enabled peer communication reduces the text-based communication barriers that impose disproportionate cognitive load on dyslexic learners, enabling natural verbal coordination without the reading and writing demands of text chat. Peer-collaborative verbal interaction has been shown to improve reading strategy use and comprehension in dyslexic children by fostering meta-cognitive awareness and peer-supported learning [16]. The non-proximity-based voice design of The Word Garden, in which players can communicate regardless of avatar positioning in the virtual world, further reduces social pressure by decoupling communication from spatial closeness in the game environment [17].

2.7 Research Gaps

Systematic review evidence consistently identifies the absence of cooperative multiplayer platforms specifically designed for dyslexic learners as a critical gap in the digital intervention landscape [4], [17]. No prior platform had combined cooperative multiplayer gameplay, dual-device cross-device literacy tasks, voice-enabled peer communication, anticipatory feedback, and a non-competitive individual reward system within a single dyslexia-focused educational experience. The Word Garden addresses this five-dimensional gap through a single integrated system, providing the first empirical and design foundation for cooperative multiplayer game-based literacy intervention for dyslexic children.

3. RESEARCH GAP ANALYSIS

The research gap addressed by The Word Garden is not a collection of individual feature absences but a single fundamental absence: no existing digital platform had ever integrated cooperative multiplayer mechanics, cross-device multimodal literacy tasks, voice-enabled peer communication, anticipatory feedback, and non-competitive individual rewards within a unified dyslexia-focused educational game. Each of these features individually has antecedents in related literature, but their integration into one coherent system targeting the cognitive, emotional, and social dimensions of dyslexia simultaneously is without precedent.

Existing dyslexia digital tools such as GraphoGame, Lexia Core5, and Nessy Learning provide evidence-based phonics practice but are universally single-player, single-device, reactive in their feedback design, and competitively framed through accuracy metrics and leaderboards. They address literacy skill remediation in isolation while ignoring the social isolation, fear of failure, and motivational barriers that research identifies as equally significant barriers to long-term learning outcomes for dyslexic children [4], [19]. The Word Garden is positioned not as a replacement for these tools but as a complementary system that addresses the dimensions they systematically neglect.

Feature	GraphoGame [4]	Lexia Core5 [10]	Nessy Learning [17]	The Word Garden
Cooperative multiplayer	X	X	X	✓
Voice-enabled communication	X	X	X	✓
Dual-device architecture	X	X	X	✓
Anticipatory feedback	X	X	X	✓
Non-competitive individual rewards	X	X	X	✓
3D letter interaction	X	X	X	✓
Private peer-invisible tasks	X	X	X	✓

Feature	GraphoGame [4]	Lexia Core5 [10]	Nessy Learning [17]	The Word Garden
Multisensory (visual+audio+tactile)	✓	✓	✓	✓
Adaptive difficulty	✓	✓	✓	✓
Competitive leaderboard	✗	✓	✓	✗ (deliberate)

Table 2: Research Gap Analysis — The Word Garden vs Established Dyslexia Intervention Platforms

4. RESEARCH PROBLEM AND OBJECTIVES

4.1 Research Problem

Children with dyslexia face a compounding set of challenges that existing digital tools systematically fail to address as an integrated whole. Single-player isolation removes peer scaffolding and normalisation of difficulty. Flat 2D text presentations amplify visual processing demands. Reactive post-error feedback reinforces fear of failure. Competitive metrics consistently foreground deficits. Text-based communication creates additional literacy barriers to social interaction. Each of these failures compounds the others, creating a learning context in which effort rarely leads to visible success and where comparison with peers consistently highlights what dyslexic learners cannot do [1], [4], [19].

The central research problem is: can an integrated cooperative multiplayer game system that simultaneously addresses the cognitive, emotional, and social dimensions of dyslexia through multimodal cross-device literacy tasks, voice-enabled peer collaboration, anticipatory feedback, and non-competitive individual rewards produce meaningful improvements in learning engagement, literacy outcomes, and socio-emotional wellbeing for dyslexic children? This question is operationalised across four integrated research dimensions that together provide empirical evidence on the system's effectiveness across the full range of its design intentions.

4.2 Main Objective

The primary objective of this research is to design, develop, and evaluate The Word Garden as an integrated cooperative multiplayer educational game system for dyslexic children aged 8 to 12 years, with the aim of improving literacy skill engagement, social confidence, motivational persistence, and cognitive accessibility through a unified multimodal, multiplayer, anticipatory, and socially supportive game-based learning environment grounded in clinical and educational evidence.

4.3 Research Dimension Objectives

Multimodal Gameplay Module

To evaluate whether the cooperative multiplayer game environment produces measurable improvements in attention, task performance, and retention in dyslexic children compared to a matched single-player condition using the same 3D world and task content, isolating the contribution of peer-collaborative multiplayer gameplay to learning outcomes.

Social Engagement System

To evaluate the impact of the voice-enabled peer communication system and the non-competitive individual reward framework on motivation, persistence, and social communication behaviours among dyslexic child participants within the multiplayer game context.

Predictive Feedback Framework

To evaluate whether the anticipatory multimodal feedback mechanism delivering visual, audio, and haptic cues before likely errors occur during mobile letter tracing tasks produces improvements in accuracy, learning retention, and motivation compared to conventional post-error feedback approaches.

Cognitive Interaction Engine

To measure and evaluate the cognitive load experienced by players during integrated multiplayer gameplay using a child-appropriate Faces Scale instrument, comparing subjective difficulty trajectories across rounds to assess whether the dual-device, cooperative game context produces manageable challenge levels for dyslexic learners.

5. SYSTEM DESIGN AND METHODOLOGY

5.1 Research Approach

This study employs a formative evaluation methodology combining iterative Agile system development, controlled technical performance evaluation, expert clinical validation, and pilot evaluation with target population participants. The methodology follows established practice in educational technology research where expert validation and technical feasibility confirmation precede empirical assessment with children, particularly when the target population comprises vulnerable learners. The four research dimensions share a single system, a single technical infrastructure, a single ethical framework, and a single pilot evaluation context — they represent analytical lenses applied to one integrated experience rather than four separate studies.

5.2 System Architecture

The Word Garden employs a distributed dual-device architecture designed to simultaneously support collaborative social learning and private individual literacy practice. The PC game component, built in Unreal Engine 5.4, manages 3D world rendering, multiplayer session management in listen-server topology via Online Subsystem Steam, 3D letter object interaction, player replication, voice communication, and reward delivery. The Android companion application, built in Flutter, handles private letter tracing task delivery at selectable difficulty levels, visual theme customisation, anticipatory feedback delivery during tracing, and cognitive load self-reporting via Faces Scale after each round.

Firebase Realtime Database serves as cloud-based middleware for all cross-device communication. The PC game uses the VaRest plugin to transmit JSON event payloads — chest discovery triggers, session state updates, and word formation events — to Firebase via REST API. The Android application maintains persistent listeners on session nodes, responds to PC-triggered events by presenting the appropriate task, and transmits only completion status flags back to Firebase, never raw performance data. This privacy-by-design approach ensures that individual literacy performance is accessible only to the child on their own device and is never shared with the multiplayer session.

Figure 1: The Word Garden System Architecture

Category	Details
PC Game Engine	Unreal Engine 5.4 (Blueprint Visual Scripting, 3D Letter Actor System, Reward Manager)
Android Application	Flutter, letter tracing interface, Faces Scale, anticipatory feedback delivery, theme selection
Middleware	Firebase Realtime Database, VaRest Plugin, JSON REST API, session node architecture
Multiplayer Network	Online Subsystem Steam, listen-server topology, multicast RPC, GameState replication
Voice Communication	Steam P2P voice via overlay, opt-in, non-proximity-based, encrypted
Feedback System	Trajectory-monitoring anticipatory feedback (visual glow, audio pitch, haptic pulse)
Reward System	Blueprint reward manager, UMG widgets, badges, stars, spoken praise, variable reinforcement
Cognitive Assessment	Child-appropriate Faces Scale (1-5 per round), structured Observation Checklist
Session Management	4-digit session codes, no persistent accounts, session-based memory reset
Concurrent Players	2 to 5 players per multiplayer session

Table 3: The Word Garden — Technologies and System Components

5.3 Gameplay Design

The Word Garden features a fantasy-themed narrative across interconnected 3D islands accessible through portals, with an Archangel character serving as in-game mentor. This narrative embeds literacy tasks within an exploratory discovery context rather than isolated drills, framing letter encounters as adventure objectives rather than test items. The design deliberately excludes competitive elements including leaderboards, comparative rankings, and accuracy-based scoring throughout all interactions.

The core gameplay loop proceeds as follows: players join a session using a four-digit code and explore the shared 3D world cooperatively. On discovering a letter chest, a Firebase event triggers a private letter tracing task on the child's paired Android device. The anticipatory

feedback system monitors the child's tracing trajectory in real time, delivering visual glow cues on the correct path, subtle audio tone shifts, and haptic pulses when the trajectory pattern indicates a likely error is approaching. Only task completion status is transmitted back to the PC on success. The chest unlocks and adds the letter to the shared collective inventory. Voice communication via Steam overlay enables free peer coordination throughout. At the central island, players discuss and arrange collected letters to form target words, completing the cooperative task and receiving individual non-competitive rewards acknowledging their effort and participation.

Figure 2: The Word Garden Gameplay Flow

5.4 Development Process

Development followed an Agile SDLC approach across six iterative phases. Requirements were gathered through consultation with clinical consultants Dr. Swarna Wijetunga and Dr. Darshani Buddhika Hettiarachchi from Lady Ridgeway Hospital, academic supervisors, and the research team. System design prioritised the integrated architecture of all four research dimensions within a single codebase, with the anticipatory feedback engine, reward manager, voice system, and cognitive assessment instruments all implemented as modular Blueprint systems within the unified Unreal Engine project.

Implementation proceeded with the PC multiplayer environment, Firebase synchronisation pipeline, and Android tracing application developed in parallel with clearly defined integration interfaces. The single-player comparison mode was implemented as a separate game mode class within the same project, replicating all game content without multiplayer or voice features to enable controlled experimental comparison for the Multimodal Gameplay research dimension. The anticipatory feedback system was implemented as a trajectory monitoring layer within the Flutter application detecting likely errors across all tracing difficulty levels. Expert validation sessions with clinical consultants refined the design across multiple iteration cycles before finalisation.

5.5 Ethical Framework and Testing

Institutional ethical approval was obtained from SLIIT's Faculty of Computing Research Ethics Board. The research team subsequently pursued approval from Lady Ridgeway Hospital's Ethics Review Committee. The Deputy Director formally declined, citing the requirement for a Clinical Trial approval from the Ministry of Health of Sri Lanka for interventional research involving government hospital patients. The team identified a private English language class serving Grade 1 and Grade 2 students as the alternative testing site, providing supervised access to children within the target age range.

A dyslexia screening instrument developed by a Consultant Child and Adolescent Psychiatrist from Lady Ridgeway Hospital was administered to all consenting students, identifying children with significant dyslexia risk indicators for inclusion. Parental and child consent was obtained through age-appropriate materials prior to all testing. Sessions ran in groups of 3 to 5 children for approximately 25 to 30 minutes under supervision of the research team and class educators.

Data collection employed four sets of research instruments, one per research dimension, all administered within the same pilot sessions. The Multimodal Gameplay evaluation used a five-section Observation Checklist with dual-column multiplayer and single-player recording, and a Post-Session Interview Card with visual preference comparison. The Social Engagement evaluation used a Parent and Teacher Feedback Form with structured behavioural observations, and a Post-Session Interview Card. The Predictive Feedback evaluation used a mode-specific Observation Checklist for anticipatory versus post-error cue responses, a standalone per-child Frustration Tally Sheet with contextual event logging, and a Post-Session Interview Card. The Cognitive Interaction evaluation used an Observation Checklist with per-round Faces Scale recording, a standalone Faces Scale display and score grid, and a Post-Session Interview Card with preference comparison.

6. SYSTEM REQUIREMENTS

6.1 Functional Requirements

- **Cooperative Multiplayer Environment:** The system shall support 2 to 5 concurrent players per session using four-digit session codes in listen-server topology with real-time 3D world state synchronisation and peer visibility.
- **3D Letter Interaction:** Letters shall be presented as interactive 3D actor objects throughout the game world, replacing flat 2D text presentation in all game interactions.
- **Dual-Device Synchronisation:** Chest discovery events shall trigger private letter tracing tasks on the paired Android application within 500ms via Firebase; only task completion status shall be returned to the PC.
- **Anticipatory Feedback:** The system shall monitor tracing trajectory patterns in real time and deliver visual, audio, and haptic anticipatory cues when a likely error trajectory is detected before the error occurs.
- **Voice Communication:** The multiplayer mode shall provide opt-in real-time voice communication via Steam overlay, non-proximity-based, active regardless of avatar positioning in the virtual environment.
- **Non-Competitive Reward System:** The system shall deliver individual effort-based rewards including badges, stars, progress indicators, and spoken praise through multicast RPC synchronisation across all clients, without rankings, leaderboards, or accuracy-based comparative metrics.
- **Cognitive Load Assessment:** The system shall support Faces Scale administration after each gameplay round with facilitator score recording across up to 10 children and 8 rounds per session.
- **Single-Player Comparison Mode:** The system shall provide a matched single-player version replicating all game content without multiplayer session management or voice communication for controlled experimental comparison.
- **Session Privacy:** Only task completion status shall be transmitted between devices; voice communication shall be handled externally by Steam infrastructure with no researcher access to voice content.

6.2 Non-Functional Requirements

- Performance: Frame rate shall remain above 30 FPS on integrated GPU hardware during active multiplayer sessions; Firebase event delivery shall not exceed 500ms latency.
- Reliability: Task delivery accuracy via Firebase shall exceed 98%; voice connectivity shall exceed 95% through Steam relay fallback; zero crash tolerance across all testing sessions.
- Scalability: The system shall support 2 to 5 concurrent players without performance degradation in reward synchronisation, voice quality, or world state consistency.
- Security: Voice transmission shall use Steam built-in encryption; all performance data shall be anonymised; no voice content shall be accessible to or stored by the research system.
- Accessibility: All mobile interfaces shall use icon-based guidance without text-reading requirements; high-contrast colour schemes conforming to dyslexia accessibility standards shall be applied throughout.
- Usability: Session joining shall be achievable within 60 seconds of device pairing; all game interactions shall be accessible to children aged 8 to 12 without prior training.

7. COMMERCIALIZATION

The Word Garden occupies a genuinely unique position in the global dyslexia EdTech market as the first platform to combine cooperative multiplayer gameplay, dual-device cross-device literacy tasks, voice-enabled peer communication, anticipatory feedback, and non-competitive individual rewards in a single integrated system. The global assistive education technology market is projected to exceed USD 4 billion by 2027, driven by increasing awareness of neurodivergent learning needs and expanding inclusive education mandates.

7.1 Market Opportunity

In Sri Lanka, the 2023 National Education Policy's mandate for inclusive digital learning tools in state schools creates immediate institutional demand. Both Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital serve as established clinical partners, providing direct access to the target population and institutional credibility for procurement. The platform's low hardware requirements — a standard Windows PC and Android smartphone — maintain accessibility across school, therapy centre, and home environments without specialist infrastructure. The modular licensing model covering institutional subscriptions, family-tier access, and premium content provides a clear path to sustainable revenue across diverse market segments including primary schools, special education units, therapy clinics, parents, and NGO-led inclusive education programmes across Sri Lanka and the wider South Asian region.

7.2 Competitive Differentiation

No existing dyslexia digital tool offers multiplayer, cooperative mechanics, dual-device support, or voice communication, let alone the complete combination present in The Word Garden. Unlike gamified EdTech platforms that employ leaderboards and competitive scoring that inadvertently increase anxiety in struggling readers, The Word Garden deliberately excludes competitive elements at every design level. The clinical partnership with LRH and SBSCH provides ecological validity and a direct pathway to real-world deployment that academic-stage competitors cannot match. The empirical evidence base generated by this

research provides institutional purchasers with peer-reviewed validation unavailable for any comparable product.

7.3 Revenue Models and Scaling

- **Institutional Licensing:** Annual subscription for schools, special education centres, and therapy practices with educator monitoring dashboards and session management tools.
- **Freemium Family Access:** Base platform access with premium game environments, expanded word banks, and enhanced reward collections available through family subscription.
- **Research Collaboration Licensing:** Academic and clinical licensing for longitudinal studies generating further validation evidence.
- **White-Label Deployment:** The integrated system licensed to government and NGO educational programmes for branded inclusive education initiatives.

Scaling pathways include a formal longitudinal clinical trial with Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital post-graduation, subject to Ministry of Health Clinical Trial approval; grant applications to the National Science Foundation of Sri Lanka and international EdTech funding bodies; and an open-source release of the design guidelines and interaction framework enabling adoption and further research globally. Future development priorities include an AI-driven adaptive engine for personalised task sequencing, iOS and tablet platform extension, and a teacher and therapist dashboard for remote progress monitoring.

8. RESULTS AND DISCUSSION

8.1 Technical Performance

Technical evaluation of The Word Garden as an integrated system was conducted over three weeks under controlled laboratory conditions with 2 to 5 concurrent players, yielding comprehensive performance data across all system dimensions.

Metric	Result	Target
Frame Rate, Discrete GPU	70-80 FPS	> 60 FPS
Frame Rate, Integrated GPU	30-35 FPS	> 30 FPS
Network Latency, 2 players	60 ms	< 150 ms
Network Latency, 5 players	150 ms	< 250 ms
Bandwidth per User, 2 players	140 kbps	< 500 kbps
Bandwidth per User, 5 players	450 kbps	< 500 kbps
Firebase Event Latency (PC-to-Android)	160-190 ms	< 500 ms
Task Delivery Accuracy	> 98.3%	> 95%
Firebase Timeout Recovery	2-4 sec (auto retry)	< 10 sec
Voice Latency, Local Network	< 100 ms	< 150 ms
Voice Latency, Internet P2P	150-250 ms	< 300 ms
Voice Connection Success Rate	> 97.3%	> 95%
Reward Sync Latency (Multicast RPC)	87 ms avg.	< 200 ms
Session Crashes, 48h continuous testing	0	0

Table 4: Technical Performance Metrics — The Word Garden Integrated System

Frame rates remained above playability thresholds across all hardware configurations. Network latency scaled predictably from 60ms at two players to 150ms at five players. Firebase event delivery exceeded 98.3% accuracy with mean latency of 160 to 190ms, well within the acceptable range for non-real-time task triggering. Voice connectivity exceeded 95% target at 97.3%, with Steam relay fallback ensuring reliable connectivity under restrictive network

conditions. Reward synchronisation averaged 87ms via multicast RPC. Zero crashes across 48 continuous testing hours confirmed system-wide architectural stability.

8.2 Expert Validation

Expert validation was conducted through iterative structured consultation with Dr. Swarna Wijetunga and Dr. Darshani Buddhika Hettiarachchi, both Consultant Child and Adolescent Psychiatrists from Lady Ridgeway Hospital, across multiple structured feedback sessions. Both experts evaluated The Word Garden as an integrated system across six dimensions on a 5-point scale, with individual scores diverging by no more than 0.5 points on any dimension, indicating strong inter-rater agreement.

Dimension	Rating (/5.0)	Expert Feedback
Cooperative Mechanics	5.0	Shared goals without competition closely align with therapeutic principles for dyslexia
Multisensory Integration	5.0	Visual, auditory, and tactile modalities across dual-device align with evidence-based approaches
Emotional Safety Design	4.5	Private mobile tasks and non-punitive feedback appropriately reduce performance anxiety
Visual Design	4.5	Dyslexia-friendly colours, high-contrast fonts, and 3D letter representation viewed favourably
Dual-Device Architecture	4.5	Effectively balances collaborative social engagement with private individual practice
Implementation Feasibility	4.0	Technically realistic; facilitator training and session duration limits are required

Table 5: Expert Validation Ratings — The Word Garden Integrated System

All six dimensions met or exceeded the 4.0 threshold, confirming that the integrated system design is clinically appropriate and therapeutically aligned for dyslexic children aged 8 to 12. Experts confirmed that the cooperative multiplayer structure without competitive elements addresses the social isolation and fear of failure that characterise the dyslexic learning experience. The dual-device architecture was recognised as effectively resolving the fundamental tension between peer collaboration and performance anxiety. Expert feedback

drove concrete design revisions including replacement of all text-based mobile instructions with icon-based guidance, increased mobile interface colour contrast, repositioning of the voice activation toggle as a persistent on-screen button, and simplification of the session-code pairing flow.

8.3 Pilot Evaluation Findings

Pilot testing was conducted at a private English language class with Grade 1 and Grade 2 students identified through the LRH psychiatrist-developed dyslexia screening instrument. Sessions of approximately 25 to 30 minutes were conducted in groups of 3 to 5 children under supervision of the research team and class educators. Findings across all four research dimensions are reported here as observations from a single integrated game experience.

Multiplayer Engagement and Attention

Children in the multiplayer condition demonstrated consistently higher on-task persistence compared to the single-player condition, with Observation Checklist Section C recording focused engagement more frequently in multiplayer sessions across all participants. Visible distraction was observed at lower frequency in the multiplayer condition, with observers noting that peer chest discovery events and voice communications recaptured wandering attention in ways the static single-player environment could not. Collaborative chest completion was observed in most multiplayer sessions, and visible excitement on chest discovery was present in both conditions, with the multiplayer context generating additional social excitement through peer awareness and shared celebration.

Social Engagement and Reward Reception

Children's responses to the non-competitive reward system were consistently positive across all sessions. Animated garden growth effects, badge unlock animations, and spoken praise statements all generated enthusiastic responses. Critically, no distress or frustration responses were observed in connection with the reward system at any point. Children who experienced difficulty with tracing tasks continued engaging without the withdrawal behaviours educators described as typical in conventional literacy contexts. Parent and Teacher Feedback Form responses confirmed elevated motivation and enjoyment, with Section B comments noting voluntary task repetition described as unusual by the class educator. Voice communication

adoption was voluntary and varied, with most children who activated voice chat using it for peer encouragement and spontaneous celebration during chest discoveries.

Anticipatory Feedback Responses

Observation Checklist responses in anticipatory feedback sessions showed that the majority of children visibly responded to at least one cue type — visual, audio, or haptic — before making an error, with the visual glow cue generating the highest observable response rate. Children in the anticipatory condition corrected their tracing trajectory after receiving cues more frequently than they abandoned the task, suggesting that the proactive guidance supported persistence rather than frustration. Frustration Tally Sheet counts were lower in anticipatory feedback sessions compared to post-error sessions, with context notes indicating that most post-error frustration events were triggered by repeated correction sequences rather than task difficulty alone. Post-session interview responses confirmed that children who noticed anticipatory cues generally described them as helpful.

Cognitive Load Observations

Faces Scale scores collected after each gameplay round showed a consistent trajectory in multiplayer sessions: higher perceived difficulty in rounds one and two, corresponding to initial unfamiliarity with the dual-device interaction flow, declining across subsequent rounds as the gameplay loop became familiar. By rounds four and five, the majority of participants were selecting Easy or Medium faces despite the same task content, suggesting that cognitive load stabilised as procedural familiarity developed. Observation Checklist Section C cognitive load indicators showed that overwhelm responses were limited to early rounds and largely resolved within the first half of each session. Children's post-session interview responses to the question of which part worked their brain hardest most frequently identified the initial device pairing and orientation rather than the letter tasks themselves.

8.4 Discussion

The Word Garden demonstrates both technical feasibility and preliminary evidence of positive engagement and motivational impact as an integrated system. The most consistent finding across all four research dimensions is that the cooperative, non-competitive, socially supportive context of the game creates a qualitatively different learning environment for dyslexic children

— one characterised by persistence through difficulty, peer normalisation of challenge, and positive emotional responses that educators consistently described as different from these children's typical responses to literacy activities.

The expert validation ratings, with two dimensions achieving perfect scores of 5.0 and all dimensions between 4.0 and 5.0, provide independent clinical confirmation that the integrated system design is therapeutically aligned across its full range of features. The technical performance results confirm that the dual-device, cloud-synchronised, cooperative multiplayer architecture is sufficiently stable for supervised small-group educational deployment, with all key metrics exceeding their targets across the full 48-hour testing period.

The ethical journey — through SLIIT approval, the LRH Deputy Director's Ministry of Health requirement, and the pivot to a private English class setting — generated important procedural knowledge regarding research governance for interventional child studies in Sri Lanka that will inform the Phase 2 clinical trial planning. The Ministry of Health Clinical Trial approval pathway is now clearly identified as the regulatory route for the planned formal longitudinal study with LRH and SBSCH.

This work constitutes Phase 1 of a longitudinal investigation, establishing the technical foundations and clinical design validation needed before proceeding to full empirical assessment of learning outcomes with the target population. Phase 2 will focus on controlled empirical evaluation of each research dimension's hypotheses with expanded participant samples, pre- and post-intervention outcome measurement, and longitudinal engagement tracking within the clinical trial framework.

9. CONCLUSION

This report presented The Word Garden, a cooperative multiplayer educational game that represents the first integrated system to combine synchronous multiplayer gameplay, dual-device cross-device literacy tasks, voice-enabled peer communication, anticipatory feedback, and a non-competitive individual reward system within a single dyslexia-focused educational game. The platform addresses dyslexia's cognitive, emotional, and social barriers simultaneously, treating the dyslexic learning experience as a multidimensional challenge that no single-feature intervention can adequately meet.

Expert validation by consultant child and adolescent psychiatrists confirmed strong alignment between the integrated system design and therapeutic principles for dyslexia intervention, with all six evaluated dimensions rated between 4.0 and 5.0 and Cooperative Mechanics and Multisensory Integration achieving perfect scores. Technical evaluation demonstrated system-wide stability with 98.3% Firebase event delivery accuracy, 97.3% voice connectivity success, predictable network scaling, and zero crashes across 48 hours. Pilot evaluation at a private English language class provided preliminary evidence of positive engagement, reduced performance anxiety, peer social support, and manageable cognitive load across all observed sessions.

The Word Garden is designed not as a replacement for validated phonics-based dyslexia interventions but as a complementary system that addresses the social, emotional, and motivational dimensions those tools systematically neglect. Its cooperative structure, non-competitive design, and clinically informed architecture position it as a foundational contribution to the evidence base for next-generation inclusive educational technology. Future work will focus on formal clinical trial evaluation with Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital, expanded word bank and curriculum coverage, AI-driven adaptive personalisation, and cross-platform deployment to reach the broadest possible population of dyslexic children.

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APPENDIX A: WORK BREAKDOWN STRUCTURE

1. Integrated System Architecture

- Define overall dual-device system architecture (PC, Firebase, Android, Steam)
- Configure Unreal Engine 5.4 with Online Subsystem Steam, VaRest plugin, Blueprint systems
- Configure Flutter development environment, Firebase project, and Android Studio
- Establish shared repository, CI/CD pipeline, and integration testing protocols

2. Core Multiplayer Environment

- Implement listen-server multiplayer topology with session code pairing
- Design and build interconnected 3D island environments with portal navigation
- Implement 3D interactive letter objects, chest discovery events, and word formation panel
- Implement player replication and world state synchronisation across clients

3. Dual-Device Synchronisation

- Implement Firebase session schema, access rules, and VaRest REST API integration
- Implement Android Flutter application with three-level letter tracing interface
- Implement visual theme customisation, difficulty selection, and Faces Scale interface
- Implement completion status transmission and PC chest unlock response pipeline

4. Social Engagement System

- Implement Steam P2P voice communication with non-proximity-based overlay activation
- Implement Blueprint reward manager with UMG badge, star, and praise delivery
- Implement variable reinforcement schedule and multicast reward synchronisation

5. Predictive Feedback Framework

- Implement real-time trajectory monitoring in Flutter tracing interface
- Implement visual glow, audio pitch shift, and haptic feedback delivery before errors

- Implement post-error comparison mode for controlled experimental evaluation

6. Cognitive Interaction Engine

- Implement Faces Scale interface on Android for per-round cognitive load recording
- Implement Observation Checklist integration for session-level cognitive load indicators

7. Single-Player Comparison Mode

- Implement separate game mode class with full task parity but no multiplayer or voice
- Validate content and structural equivalence across multiplayer and single-player modes

8. Accessibility and Evaluation Infrastructure

- Implement icon-based guidance throughout mobile interface
- Implement dyslexia-friendly colour and typography standards across all UI layers
- Produce all data collection instruments for four research dimensions
- Implement session event logging with timestamps for research analysis

9. Testing, Validation, and Dissemination

- Technical testing across all system components and integration interfaces
- Expert validation consultation with LRH consultant psychiatrists
- Dyslexia screening, consent, and pilot evaluation at private English class
- IEEE conference paper, individual component final reports, and group final report

APPENDIX B: PROJECT GANTT CHART

[illegible]

[illegible]

Task	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Final Viva Presentation												**

Table 6: Project Gantt Chart — The Word Garden (July 2025 to June 2026)

APPENDIX C: BUDGET JUSTIFICATION

Requirement	Approx. Cost (LKR)
Development and Testing Resources (Firebase, Unreal Marketplace, hosting)	20,000.00
Child-Safe Headsets for Testing Sessions (3 units)	24,000.00
Participant Incentives (certificates, tokens of appreciation)	10,000.00
Printing and Research Materials (consent forms, checklists, Faces Scale)	5,000.00
Internet and Server Costs (Firebase, Steam App ID)	5,000.00
Travelling Costs (site visits, testing logistics)	5,000.00
Ethical Approval Administration	2,000.00
Publishing Costs (conference and journal submission fees)	15,000.00
Consultant Facilitation	10,000.00
Contingency and Miscellaneous	5,000.00
TOTAL	101,000.00

Table 7: Project Budget Justification